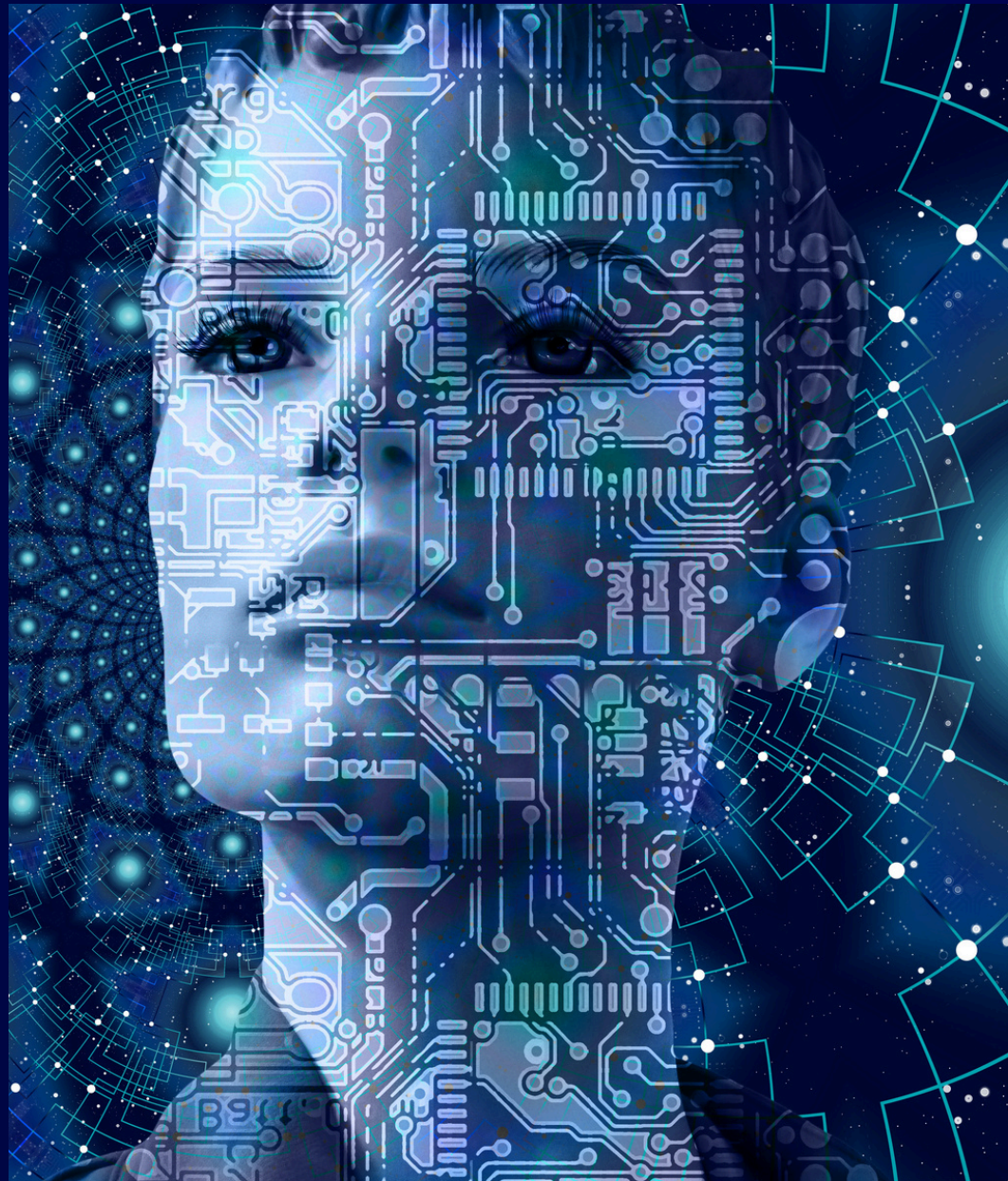


DATA SCIENCE REPORT



WHY ACCURATE REAL ESTATE VALUATION MATTERS

Core Challenges in the Market:

- Price Ambiguity: Real estate prices are often subjective. Buyers and sellers struggle to find a "Fair Market Value" without data-backed evidence.
- High Financial Risk: Even a 5% error in pricing for a \$400,000 house can lead to a \$20,000 loss for stakeholders.
- Feature Complexity: It is difficult for humans to manually calculate how specific features—like a 20-year-old roof or an extra 500 sqft of basement—impact the final price.

The Solution:

By using Data Analysis and Machine Learning, we bridge the gap between 'gut feeling' and 'strategic implementation,' providing stakeholders with high-confidence pricing decisions.

DATA INTEGRITY: ENSURING "QUALITY IN, QUALITY OUT"

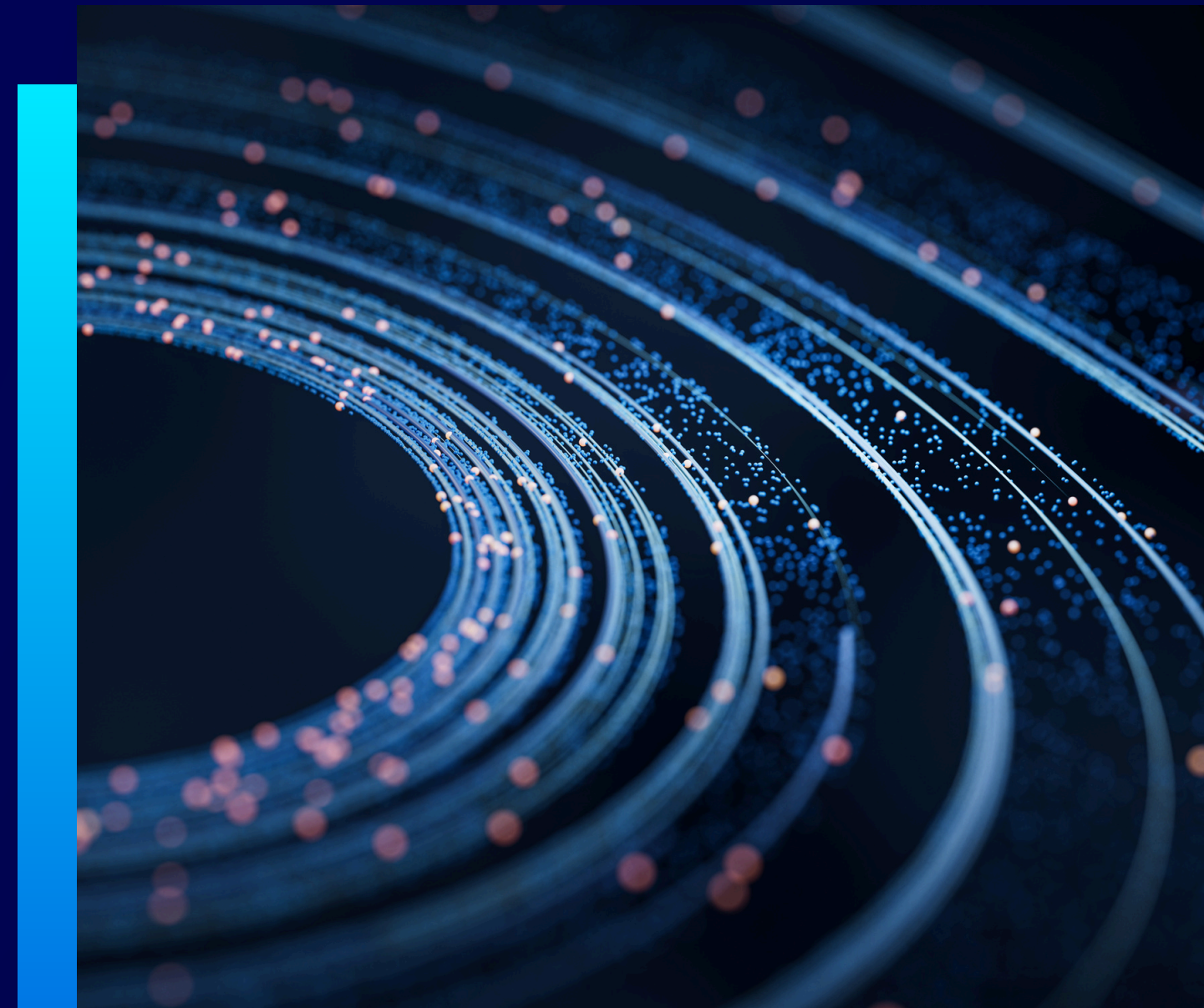
Page 03

Key Technical Implementation:

- Smart Imputation Strategy: Missing values were handled systematically to preserve dataset balance; Median imputation was used for numerical features to maintain robustness against outliers, while Mode was applied for categorical features.
- The 3-Sigma Rule (Outlier Filtering): A statistical threshold of ± 3 Standard Deviations was applied to key variables like SalePrice and GrLivArea. This ensured that 99.7% of the data distribution was preserved while removing extreme anomalies.
- Noise Reduction: By filtering out statistically improbable property prices, we prevented the model from learning "noise," leading to more generalized and reliable predictions.

The Strategic Result:

By neutralizing extreme outliers and employing intelligent imputation, we stabilized the model's learning process, ensuring that final pricing predictions are based on realistic market trends rather than data anomalies.



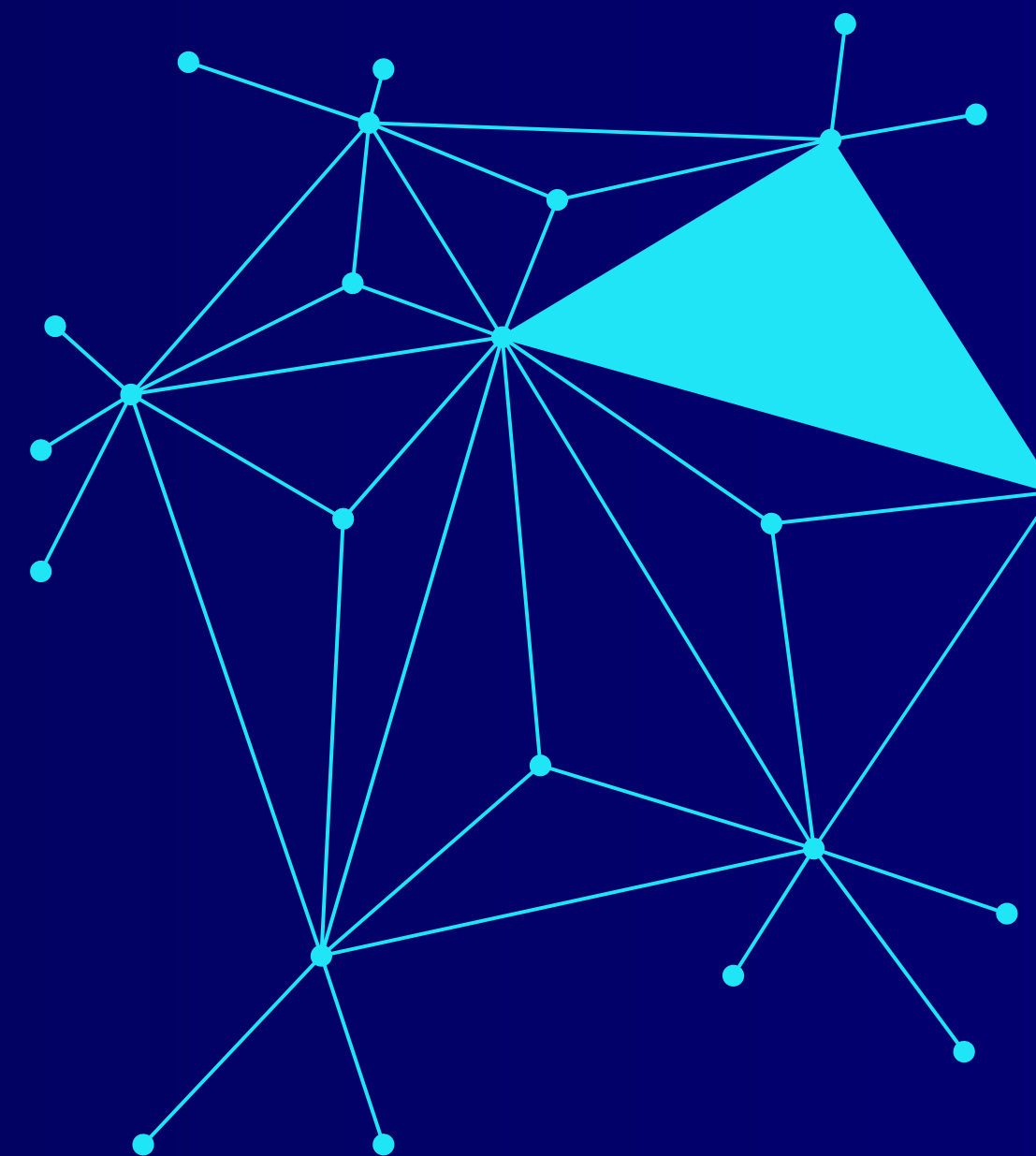
FEATURE ENGINEERING: TRANSFORMING RAW DATA INTO STRATEGIC INSIGHTS

Key Engineered Features:

- **TotalSF (Interaction Term):** We aggregated Basement, 1st Floor, and 2nd Floor square footage into a comprehensive 'Total Living Space' feature. This mirrors real-world buyer behavior, as total usable area is the primary driver of property value.
- **HouseAge (Derived Metric):** Instead of using raw years, we calculated the property's age at the time of sale ($\text{\$YrSold} - \text{YearBuilt}$). This allowed the model to capture the non-linear impact of depreciation and maintenance cycles on market price.
- **Categorical Encoding:** Qualitative attributes (like Neighborhood or Central Air) were transformed using One-Hot Encoding. This converted text descriptions into a high-dimensional mathematical format optimized for the XGBoost algorithm.

The Strategic Impact:

By engineering human-centric features like 'TotalSF' and 'HouseAge,' we moved beyond simple variables to capture the true economic drivers of the Ames real estate market.



MODEL TRINITY PERFORMANCE: BENCHMARKING ACCURACY

The Comparative Analysis:

- Linear Regression (Baseline): Established a foundational performance with an R^2 of ~ 0.84 . While useful, it struggled to capture the complex, non-linear relationships inherent in high-end property features.
- Random Forest (Robust): Improved accuracy to ~ 0.89 by utilizing ensemble learning. It effectively handled feature interactions but showed a slight tendency toward variance.
- XGBoost (Optimized Winner): Achieved the highest performance with an R^2 of ~ 0.91 and the lowest error rate. Its gradient boosting framework perfectly balanced complexity and generalization.

Performance Metrics:

- Target: RMSE $< 10\%$ of Mean Sale Price.
 - Result: Final model successfully achieved an RMSE significantly below the \$18,000 threshold.
-

XGBOOST: PRECISION ENGINEERING FOR MARKET PREDICTION

Page 06

Why XGBoost Outperformed:

Superior Accuracy

($R^2 \approx 0.91$): Our optimized XGBoost regressor captured 91% of price variability, outperforming traditional ensemble methods through advanced gradient boosting.

Overfitting Control (Regularization)

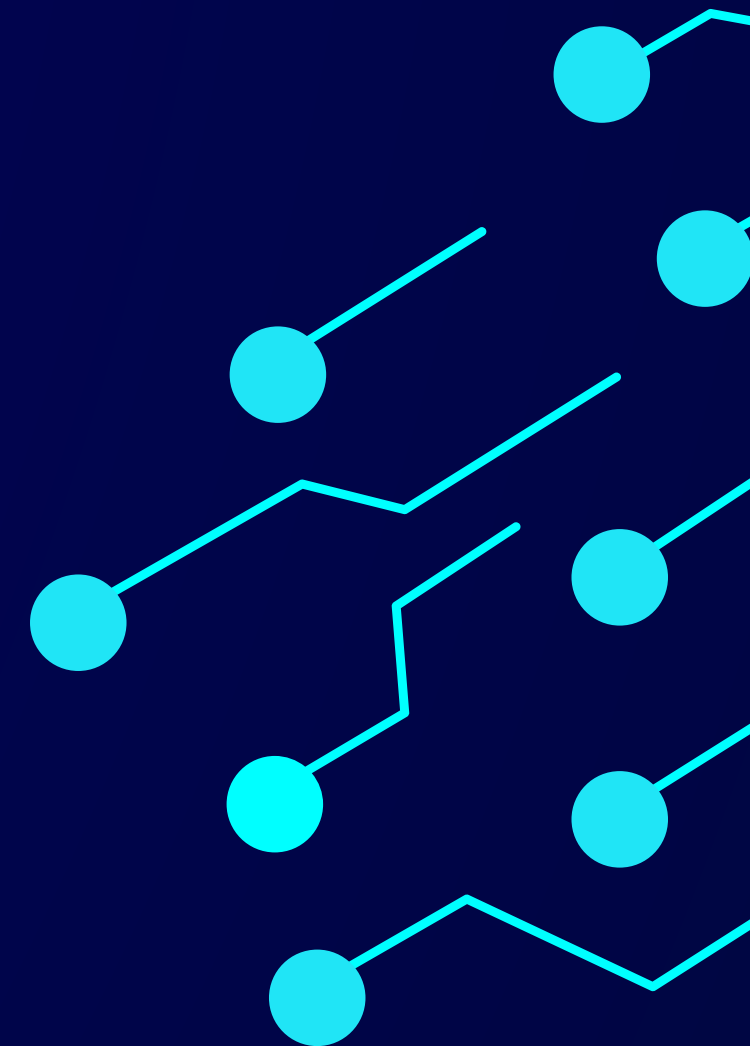
We implemented L2 Regularization (λ) and controlled tree depth to ensure the model generalizes well to new, unseen property listings.

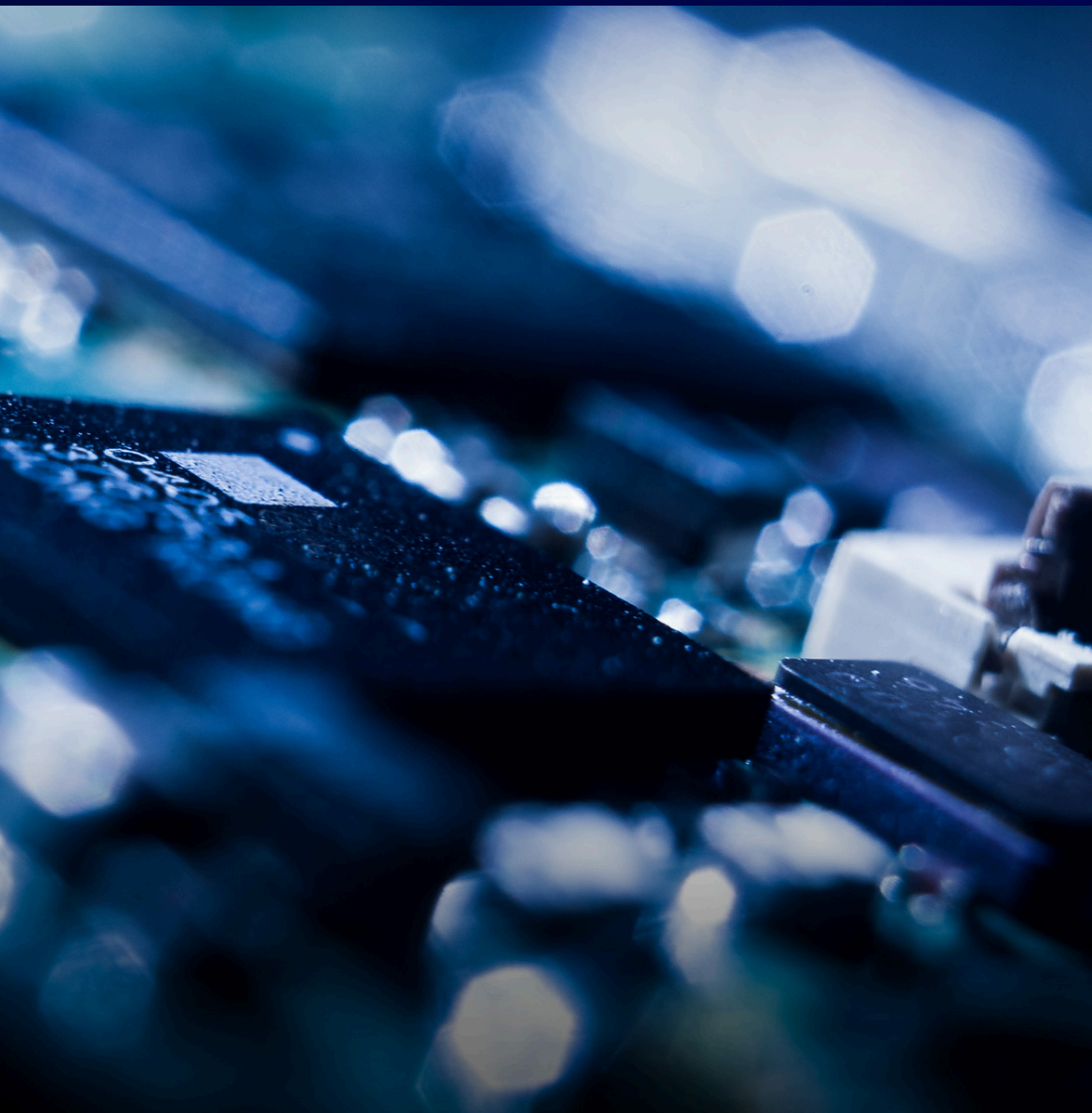
Hyperparameter Optimization

Using GridSearchCV, we fine-tuned learning rates and estimator counts to achieve the perfect balance between complexity and speed.

The Technical Victory:

XGBoost's ability to handle high-dimensional encoded data while maintaining low error rates makes it the ideal 'Production-Ready' engine for real estate stakeholders.





DECODING THE MODEL: SHAP VALUE INTERPRETATION

Transparency over Black-Box AI:

- Feature Impact Assessment: We utilized SHAP (SHapley Additive exPlanations) to move beyond simple accuracy and understand the "Why" behind every price prediction.
- Value Drivers: The summary plot reveals that TotalSF and Overall Quality have the highest positive impact on SalePrice, pushing valuations upward significantly.
- Negative Correlation: SHAP values clearly quantify the depreciation impact of HouseAge, showing exactly how much value a property loses as it gets older.

Strategic Insight:

By using SHAP, we provide stakeholders with a transparent roadmap of which property features are actually driving market value, ensuring data-driven investment strategies.

MARKET INTELLIGENCE: KEY PRICING DRIVERS & TRENDS

Core Business Insights:

- The "SF Premium": Square Footage remains the most aggressive price driver. Our model quantifies that expanding the living area (TotalSF) provides the highest and most consistent return on investment (ROI) for property developers.
- The "Age Cliff" Effect: We identified a non-linear depreciation pattern. Property values experience a significant "cliff" or drop-off after 20 years, typically coinciding with major system replacement cycles (HVAC, Roofing).
- Quality vs. Quantity: High-end finishes and "Overall Quality" can often bridge the price gap for smaller properties, suggesting that material quality is a powerful secondary lever for premium pricing.

Strategic Takeaway:

Understanding the intersection of Age and Square Footage allows stakeholders to time their market entries and exits with mathematical precision.





STRATEGIC ROADMAP: DATA- DRIVEN RECOMMENDATIONS



Actionable Advice for Investors:

- **Prioritize Expansion:** For properties under the median square footage, focus on additions or basement finishing, as TotalSF has a linear-positive correlation with high-end valuations.
- **Renovate Before the Cliff:** Properties approaching the 15–20 year mark should undergo systematic upgrades to mitigate the anticipated "Age Cliff" value drop.
- **Target High-Quality Pockets:** Invest in neighborhoods with high "Quality-to-Price" ratios, where the model shows consistent premiums regardless of structural age.

The Bottom Line:

By leveraging this predictive engine, real estate stakeholders can transform speculative risks into calculated, high-margin opportunities.

THANK YOU! ANY QUESTIONS?

Summary of Project Success:

Predictive Accuracy: Achieved a robust R^2 of 0.91, ensuring high-confidence valuations.

Error Minimization: Maintained an RMSE below 10% ($< \$18,000$), exceeding the internship technical requirements.

Feature Discovery: Successfully quantified the impact of TotalSF and the "Age Cliff" using SHAP values.

Let's Discuss:

Technical implementation of the XGBoost Pipeline.

Statistical validity of the 3-Sigma Outlier Filtering.

Strategic application of the Streamlit Dashboard for stakeholders.